A/B Testing and Its Importance in Business Decisions

Created By: Preshna Acharya

1. What is A/B testing?

A/B testing is a method used to compare two versions of something to see which works better. Version A is usually the current version, while Version B is a new or changed version. People are randomly assigned to one version, and their results are compared.

Examples: two website buttons, two advertisement designs, or two signup pages.

2. Simple example: an online store

An online store wants more customers to complete a purchase. It tests its old "Buy Now" button (A) against a new button design (B). Each version is shown to a randomly selected group of 1,000 customers.

Measure	Version A (old)	Version B (new)
Visitors	1,000	1,000
Customers who purchase	100	130
Conversion rate	10%	13%

Conversion rate = (number of customers who purchase ÷ number of visitors) × 100.

What happened? Version B had a 13% conversion rate compared with 10% for A. That is 3 percentage points higher (30% higher relative to A). This is an encouraging result, but a statistical test is needed to check whether the difference is likely to be random variation.

3. Why is A/B testing important for businesses?

- **Better decisions:** Companies use evidence instead of guesses.
- **More sales or sign-ups:** They can find a website or message that works better.
- **Better customer experience:** They can test what customers find easier or more useful.
- **Lower risk:** A new idea can be tested with limited exposure before a wider launch.
- **Clear measurement:** Teams can track a chosen outcome, such as purchase rate or revenue.

4. How to perform an A/B test

1. Choose one business goal, such as improving purchase rate.
2. Create versions A and B and decide what success means.
3. Randomly assign similar visitors to A or B.
4. Collect enough data over an appropriate period.
5. Compare results using a statistical test and check for negative effects.
6. Choose whether to keep, reject, or test the change again.

5. Connection with hypothesis testing

Null hypothesis (H_0): Version A and Version B have the same true conversion rate.

Alternative hypothesis (H_1): The two versions have different true conversion rates.

A two-proportion z-test or chi-square test can compare purchase rates. A p-value below 0.05 is commonly called statistically significant, but it does not guarantee the change is valuable or that future results will be the same.

For example: an uncorrected chi-square test gives $p \approx 0.0355$. This is statistically significant at the 5% level, assuming independent customers and a correctly randomized experiment.

6. Important points to remember

A/B testing is not just comparing two existing groups. Random assignment helps make the comparison fair. Other business metrics such as refunds, website speed, profit, and customer complaints should also be monitored. Statistical significance does not automatically mean a large or profitable improvement.

7. Conclusion

A/B testing helps businesses compare two versions using customer behavior and data. It can guide improvements in websites, advertisements, and products. A well-designed test helps businesses make more reliable decisions instead of changing things based only on opinions.